

Shaged Carasso, Ph.D.

Postdoctoral Associate | Xavier Lab, Broad Institute
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Research Experience

- 08/2025 - Present **Postdoctoral Associate**
Broad Institute of MIT and Harvard, Cambridge, MA, USA
Advisor: Prof. Ramnik Xavier
- 06/2021 – 07/2025 **Doctorate in Medical Sciences (Bioinformatics, microbiology)**
Technion – Israel Institute of Technology, Haifa, Israel
Main Advisor: Associate Professor Naama Geva-Zatorsky
Associate Advisor: Dr. Itai Sharon (MIGAL Galilee Research Institute)
- 10/2018 – 05/2021 **Magistrate in Medical Sciences**
Technion – Israel Institute of Technology, Haifa, Israel
Advisor: Associate Professor Naama Geva-Zatorsky

Education

- 06/2021 – 07/2025 **Technion – Israel Institute of Technology, Haifa, Israel**
Ph.D., Bioinformatics
Advisors: Naama Geva-Zatorsky, Itai Sharon
Thesis: Phase Variations of the Bacteroides Genus
- 10/2018 – 05/2021 **Technion - Israel Institute of Technology, Haifa, Israel**
M.Sc., Medical Sciences (Cum Laude)
Advisor: Naama Geva-Zatorsky
Thesis: Fecal Microbiota Transplantation and Phage Therapy to Eradicate Antibiotic Resistant Bacteria
- 10/2014 – 10/2018 **Technion - Israel Institute of Technology Haifa, Israel**
B.Sc., Biology

Teaching

- 10/2021 – 03/2025 **Teaching Assistant**
Technion - Israel Institute of Technology, Haifa, Israel
Medical Bacteriology Lab (Undergraduate Medical School Course)

Publications

Published Papers

- Ohayon C, Gross N, Capucha T, **Carasso S**, Shkedy Y, Rachmiel A, Emodi O. Efficiency and outcomes in microvascular anastomosis: A meta-analysis of mechanical versus manual techniques. *Journal of Cranio-Maxillofacial Surgery*. July 2025.
- Ohayon C, Ritschl LM, Capucha T, **Carasso S**, Fichter AM, Gross N, Rachmiel A, Wolff KD, Emodi O. The never-ending story: single versus dual venous anastomoses in microsurgical radial forearm free flap for head and neck reconstruction. *Journal of Cranio-Maxillofacial Surgery*. June 2025.
- #Naddaf R, #Shmilovich H, **#Carasso S**, Keshet-David R, Herren R, Gefen T, et al. Esophageal microbiome correlates with post-esophagectomy anastomotic leak in cancer patients. *ESMO Gastrointestinal Oncology*. June 2025. (# Equal Contribution)
- Gal-Mandelbaum N, **Carasso S**, Kedem A, Ziv T, Keshet-David R, Abboud R, et al. Dietary carbohydrates alter immune-modulatory functionalities and DNA inversions in *Bacteroides thetaiotaomicron*. *Nature Communications*. May 2025.
- #Carasso S**, #Keshet-David R, Zhang J, Hajjo H, Kadosh-Kariti D, Gefen T, et al. Bacteriophage-driven DNA inversions shape bacterial functionality and long-term co-existence in *Bacteroides fragilis*. *Gut Microbes*. 2025. (# Equal Contribution)
- #Carasso S**, #Zaatry R, #Hajjo H, #Kadosh-Kariti D, Ben-Assa N, Naddaf R, et al. Inflammation and bacteriophages affect DNA inversion states and functionality of the gut microbiota. *Cell Host & Microbe*. March 2024. (# Equal Contribution)

7. #Naddaf R, #Carasso S, Reznick-Levi G, Hasnis E, Qarawani A, Maza I, et al. Gut microbial signatures are associated with Lynch syndrome (LS) and cancer history in Druze communities in Israel. *Scientific Reports*. Nov 2023. (# Equal Contribution)
8. Mandelbaum N, Zhang L, Carasso S, Ziv T, Lifshiz-Simon S, Davidovich I, et al. Extracellular vesicles of the Gram-positive gut symbiont *Bifidobacterium longum* induce immune-modulatory, anti-inflammatory effects. *npj Biofilms and Microbiomes*. 2023.
9. #Hadar Mudrik-Zohar H, #Carasso S, Gefen T, Zalmanovich A, Katzir M, Cohen Y, et al. Microbiome Characterization of Infected Diabetic Foot Ulcers in Association with Clinical Outcomes: Traditional Cultures versus Molecular Sequencing Methods. *Frontiers in Cellular and Infection Microbiology*. 2022. (# Equal Contribution)
10. Carasso S, Fishman B, Lask LS, Shochat T, Geva-Zatorsky N, Tauber E. Metagenomic analysis reveals the signature of gut microbiota associated with human chronotypes. *The FASEB Journal*. 2021.
11. Marai I, Carasso D, Carasso S, Carasso S. The Mechanical Cost of Decreasing Conduction Velocity: A Mathematical Model of Pacing-Induced Lower Strain. *Journal of Atrial Fibrillation*. 2021.
12. Berkowitz E, Kopelman Y, Kadosh D, Carasso S, Tiosano B, Kesler A, Geva-Zatorsky N. "More Guts than Brains?"—The Role of Gut Microbiota in Idiopathic Intracranial Hypertension. *Journal of Neuro-Ophthalmology*. 2021.
13. #Bar-Yoseph H, #Carasso S, Shklar S, Korytny A, Even Dar R, Daoud H, et al. Oral Capsulized Fecal Microbiota Transplantation for Eradication of Carbapenemase-producing Enterobacteriaceae Colonization With a Metagenomic Perspective. *Clinical Infectious Diseases*. June 2020. (# Equal Contribution)
14. Ben-Assa N, Coyne MJ, Fomenkov A, Livny J, Robins WP, Muniesa M, Carey V, Carasso S, et al. Analysis of a phase-variable restriction modification system of the human gut symbiont *Bacteroides fragilis*. *Nucleic Acids Research*. 2020.

Previews

15. Carasso S, Hajjo H, Geva-Zatorsky N. Phage-Bacteria Associations: Analyze. Match. Develop Therapies. *Cell Host & Microbe*. 2020.

Patents

1. Geva-Zatorsky N, Carasso S, Gefen T, Keshet R, Hajjo H, Zaatry R, Sharon I, Reversible bacterial phase-variations and uses thereof in diagnostic and therapeutic applications (pending), WO2024176230A1. World Intellectual Property Organization; 2024.
2. Geva-Zatorsky N, Gefen T, Kariti D, Ben-Assa N, Abboud R, Carasso S, Zaatry R, Bacteriophages as a tool to manipulate gut commensal immune-modulation activity (pending), WO2024194876A1. World Intellectual Property Organization; 2024.

Awards and Honors

2025	Best Lecture Award – The Technion’s Faculty of Medicine Graduate Research Day.
2024	Jacobs Award for an Excellent Publication (Technion award for best publication).
2024	Best Poster Award – The Technion’s Faculty of Medicine Graduate Research Day.
2023	Gutwirth Excellence Scholarship.
2022	Teva’s BioInnovation Student Fellowship & Mentorship Program.
2020-24	Faculty of Medicine Excellence Scholarship for Graduate Students (Awarded five semesters).
2017	Outstanding Bachelor Student Award, Faculty of Biology.
2017-18	Faculty of Biology Dean’s List (Awarded three semesters).

Conference Presentations/Posters

2024	“Bacterial DNA inversions and functional plasticity during inflammation and encounter with bacteriophages”, Israel Society for Microbiology 2024 Annual Meeting, Bar Ilan University, Ramat Gan, Israel (poster)
2024	“Long-term impact: bacteriophage-driven DNA inversions shape <i>Bacteroides fragilis</i> functionality”, 10th International Human Microbiome Consortium Congress (IHMC), Rome, Italy (short presentation)
2024	“Bacterial DNA inversions and functional plasticity during inflammation and encounter with bacteriophages”, Jacobs Graduate Research Day, Technion, Haifa, Israel (poster)

- 2023 “Bacterial DNA inversions and functional plasticity during inflammation and encounter with bacteriophages”, EMBL Symposium The human microbiome, Heidelberg, Germany (presentation)
- 2023 “Inflammation and bacteriophages affect DNA inversion states and functionality of the gut microbiota”, 10th Annual Broad-ISF Symposium, Jerusalem, Israel (poster)
- 2023 “Gut Bacteroides exhibit genomic phase variations under inflammatory conditions”, ILANIT | FISEB Experimental Biology Conference, Eilat, Israel (presentation)